

DATASET DESCRIPTION

This document describes the two public road-extraction datasets used to train and evaluate DBR-Net: the Massachusetts Roads Dataset and the DeepGlobe Road Extraction Dataset. Both datasets formulate road extraction as binary semantic segmentation, but they differ in spatial resolution, geographical coverage, scene composition, and annotation characteristics.

1 Massachusetts Roads Dataset

Dataset source: A public aerial-image dataset introduced by Volodymyr Mnih in 2013 and distributed through the University of Toronto. The copy used in this study was obtained from the Kaggle mirror listed in Section 4.

1.1 Overview of the Dataset

The Massachusetts Roads Dataset is a widely used benchmark for extracting road networks from high-resolution aerial imagery. It covers urban, suburban, and rural areas across Massachusetts, USA. Each input image is paired with a binary target map in which road pixels form the foreground class and all remaining pixels form the background class. The large native images and the mixture of dense street networks, major roads, and narrow local roads make the dataset suitable for evaluating both regional road coverage and fine-structure recovery.

1.2 Dataset Composition

- **Total size:** 1,171 aerial image-mask pairs.
- **Official training split:** 1,108 image-mask pairs.
- **Official validation split:** 14 image-mask pairs.
- **Official test split:** 49 image-mask pairs.
- **Image size:** 1500×1500 pixels.
- **Spatial resolution:** approximately 1 m/pixel.
- **Geographical coverage:** more than 2,600 km².
- **Data type:** three-channel optical aerial imagery with binary road target maps generated from vector road data.
- **Task:** pixel-level binary semantic segmentation of roads and background.

1.3 Image and Annotation Characteristics

- **Scene diversity:** The images include dense residential areas, suburban neighbourhoods, highways, intersections, vegetation, water bodies, and sparsely developed regions.
- **Road morphology:** Road networks contain long linear segments, curved streets, junctions, loops, and narrow branches at multiple orientations and scales.
- **Occlusion and ambiguity:** Trees, buildings, shadows, and vehicles can obscure road surfaces, while rivers, roofs, and other elongated structures may resemble roads locally.
- **Class imbalance:** Road pixels occupy a substantially smaller image area than background pixels.

- **Native-image challenge:** The 1500×1500 resolution requires a model to preserve fine details while retaining sufficient contextual information over a large field of view.

1.4 Use in the Present Study

- **Benchmark role:** Massachusetts Roads serves as the first benchmark for training and evaluating DBR-Net and the comparison models.
- **Training input:** Training uses random 1024×1024 crops from the native images.
- **Data augmentation:** Random horizontal and vertical flips, rotations by 90° , 180° , or 270° , and random scaling with a factor sampled from $[0.75, 1.0]$ are applied.
- **Centerline supervision:** Auxiliary centerline targets are generated from the binary masks by tensor-based morphological thinning before downsampling; they are not separate annotations supplied by the dataset.
- **Evaluation:** Predictions are binarised at a threshold of 0.5. Precision, recall, F1-score, and road-class intersection over union (IoU) are computed from pixel-level confusion statistics.

1.5 Dataset Strengths and Limitations

- **Strengths:** The dataset provides public pixel-level targets, large native images, varied road layouts, and a standard split that supports comparison with previous road-extraction studies.
- **Limitations:** Its geographical coverage is restricted to one US state, and the rasterized road labels may not represent the exact visible width of every road surface. Results on this dataset alone therefore do not establish generalisation to other countries, sensors, or annotation conventions.

2 DeepGlobe Road Extraction Dataset

Dataset source: The road-extraction track of the DeepGlobe 2018 Satellite Image Understanding Challenge. The copy used in this study was obtained from the Kaggle mirror listed in Section 4.

2.1 Overview of the Dataset

The DeepGlobe Road Extraction Dataset was developed for large-scale road segmentation from satellite imagery. It contains geographically diverse scenes from Thailand, Indonesia, and India and includes urban, suburban, rural, and agricultural environments. Unlike centerline-only annotations, its binary masks label the visible road area at the pixel level. The variation in road material, density, surrounding land cover, and illumination makes DeepGlobe a challenging benchmark for testing model robustness across heterogeneous scenes.

2.2 Dataset Composition

- **Total size:** 8,570 RGB satellite images.
- **Original challenge split:** 6,226 training images, 1,243 validation images, and 1,101 test images.

- **Publicly labelled subset:** 6,226 training image-mask pairs in the commonly distributed release; the original validation and test labels are not included in that release.
- **Image size:** 1024×1024 pixels.
- **Spatial resolution:** approximately 0.5 m/pixel.
- **Geographical coverage:** approximately 2,220 km².
- **Data type:** three-channel RGB satellite imagery with binary pixel-level road masks.
- **Class distribution:** Road pixels account for approximately 4.5% of the original training split, illustrating the strong foreground imbalance of the task.

2.3 Image and Annotation Characteristics

- **Geographical diversity:** Samples cover multiple countries and include urban, suburban, rural, agricultural, forested, and coastal environments.
- **Road diversity:** The dataset contains paved and unpaved roads, highways, local streets, intersections, and road networks with widely varying density and width.
- **Pixel-level road areas:** The masks delineate road surfaces rather than only their centerlines, enabling direct road-area segmentation.
- **Challenging appearance:** Illumination changes, vegetation, shadows, low-contrast unpaved roads, and visually similar non-road structures increase ambiguity.
- **Foreground imbalance:** Most pixels belong to the background, making foreground-sensitive metrics such as road IoU and F1-score more informative than overall pixel accuracy alone.

2.4 Use in the Present Study

- **Benchmark role:** DeepGlobe serves as the second benchmark for evaluating DBR-Net under broader geographical and scene variation.
- **Reproducible labelled split:** The 6,226 labelled pairs are shuffled with seed 3407. Exactly 5,000 samples are used for training and the remaining 1,226 samples form the full test holdout. The first 300 samples of this holdout are also used for validation and remain within the reported 1,226-image test set.
- **Model-selection dependency:** Because the 300-image validation subset is contained in the test holdout, the DeepGlobe test score is not fully independent of model selection. This dependency must be considered when interpreting the result.
- **Training input:** The native image size is 1024×1024 pixels, matching the training crop size used in the main experiments.
- **Training and evaluation:** The same augmentation operations, optimisation policy, decision threshold, and pixel-level metrics are applied to DBR-Net and all comparison models.

2.5 Dataset Strengths and Limitations

- **Strengths:** DeepGlobe offers large-scale, geographically varied satellite imagery and pixel-level road-area annotations covering diverse road surfaces and scene types.
- **Limitations:** Public labels are concentrated in the original training portion, which requires a custom split for reproducible local evaluation. The present study’s overlapping validation and test protocol also introduces a model-selection dependency.

3 Comparative Summary

Property	Massachusetts Roads	DeepGlobe Roads
Image modality	RGB aerial imagery	RGB satellite imagery
Native size	1500 × 1500 pixels	1024 × 1024 pixels
Spatial resolution	Approximately 1 m/pixel	Approximately 0.5 m/pixel
Total images	1,171	8,570
Geographical scope	Massachusetts, USA	Thailand, Indonesia, and India
Road annotation	Binary rasterized road target maps	Binary road-surface masks
Primary challenge	Large scenes and fine road structures	Cross-region and appearance diversity

Together, the datasets provide complementary evaluation conditions. The Massachusetts Roads Dataset tests extraction from large aerial tiles within a consistent regional domain, whereas DeepGlobe tests robustness to broader geographical, environmental, and road-surface variation.

4 Data Availability

The dataset copies used in this study are publicly accessible through Kaggle:

- Massachusetts Roads Dataset:
<https://www.kaggle.com/datasets/balraj98/massachusetts-roads-dataset>
- DeepGlobe Road Extraction Dataset:
<https://www.kaggle.com/datasets/balraj98/deepglobe-road-extraction-dataset>

The original Massachusetts Roads Dataset and its citation information are available from the University of Toronto at <https://www.cs.utoronto.ca/~vmnih/data/>. The DeepGlobe dataset is described in I. Demir et al., “DeepGlobe 2018: A Challenge to Parse the Earth Through Satellite Images,” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2018, pp. 172–181, https://openaccess.thecvf.com/content_cvpr_2018_workshops/html/Demir_DeepGlobe_2018_A_CVPR_2018_paper.html.